Machine Learning Systems – Report Marking Scheme (2025/26)

INFR11269 Machine Learning Systems · Proposed 2025/26 Marking Scheme

Designed to align with the 2025/26 report template and with the University of Edinburgh Common Marking Scheme style of descriptors.

Purpose and marking principles

- The report counts for 70% of the coursework mark. The report and supplementary code should be judged together, but marks for writing quality should be based on what is presented in the report itself.
- The report must be assessed against the 2025/26 template: Introduction, Implementation, Performance Profiling, Bottleneck Analysis, Optimization Attempts, Comparison, and Conclusion. Credit should follow the quality of evidence and reasoning, not merely the presence of headings.
- Markers are not required to read appendices. The main body should stand on its own.

Grade bands and descriptor style

Use the University of Edinburgh Common Marking Scheme bands when turning criterion-level judgements into an overall mark: A1/A2/A3 = 70–100, B = 60–69, C = 50–59, D = 40–49, Fail = below 40. Within each band, markers should reward evidence, technical accuracy, depth of analysis, and clarity of communication rather than simple section completion.

Band	Indicative meaning	How it should look on this coursework
A1/A2/A3 (70–100)	Excellent	Technically correct, analytically strong, evidence-led, and communicated with clarity and precision.
B (60–69)	Very good	Mostly strong technically, with good evidence and clear structure, but with some gaps in justification, depth, or interpretation.
C (50–59)	Good	Meets the brief at a basic level, but analysis is descriptive, evidence is thinner, or important technical choices are under-justified.
D (40–49)	Satisfactory / borderline	Partial coverage only; important errors, weak evidence, or limited understanding reduce confidence in the work.

Fail (<40)	Unsatisfactory	Serious omissions, major misunderstandings, or inadequate evidence prevent the report from meeting course expectations.

Section weights

Section	Weight	What markers should look for
1. Introduction	10%	Clear ASR/GLM-ASR motivation, accurate end-to-end pipeline description, sensible overview of kernels, track, optimizations, and headline findings.
2. Implementation	18%	Well-justified kernel design, phase-by-phase implementation summary, thoughtful discussion of memory-access patterns, data layout, and parameter choices.
3. Performance Profiling	14%	Sound benchmarking setup, appropriate metrics, and convincing end-to-end plus operator-level profiling evidence.
4. Bottleneck Analysis	14%	Correct use of arithmetic intensity, bandwidth/TFLOPS reasoning, and a persuasive diagnosis of the true overall bottleneck.
5. Optimization Attempts	18%	At least one optimisation analysed through Hypothesis → Change → Result, with technically credible interpretation of success or failure.
6. Comparison	16%	Fair comparison with baseline, good data collection, clear per-operator and end-to-end results, and convincing root-cause analysis.
7. Conclusion and scholarly communication	10%	Clear synthesis of findings, realistic next steps, concise reflection on lessons learned, and professional presentation throughout.

Detailed rubric by section

Criterion	Weight	A (70–100)	B (60–69)	C (50–59)	D/Fail (<50)
1. Introduction	10%	Motivation is accurate and specific to ASR inference; the pipeline is clearly explained; the system overview names the track and kernels; headline findings are informative rather than generic.	Mostly accurate and clear, but some links between motivation, kernels, and findings are brief or underdeveloped.	Covers the requested points, but mainly at a descriptive level; terminology or rationale may be shaky in places.	Missing key context, technically inaccurate, or too vague to orient the reader.
2. Implementation	18%	Explains the five workflow phases clearly; distinguishes compute- vs bandwidth-oriented behaviour; justifies tile/block sizes, warps/stages, tensor layout, and GQA mapping with sound technical reasoning.	Implementation is clear overall, but some choices are asserted rather than justified, or the memory-access discussion lacks precision.	Basic implementation summary is present, but design choices are only lightly justified and technical detail is uneven.	Implementation is incomplete, confused, or unsupported by technical explanation.
3. Performance Profiling	14%	Benchmark setup is reproducible and fair; metrics are appropriate; profiling evidence is well organised; end-to-end and component-level results are both informative.	Good measurement practice overall, though some controls, baselines, or profiling details could be stronger.	Some evidence is presented, but the measurement design is weak, thin, or poorly explained.	Profiling is missing, unreliable, or too weak to support later claims.
4. Bottleneck Analysis	14%	Uses GPU specs, arithmetic intensity, and kernel evidence accurately; clearly separates kernel bottlenecks from launch/sync/allocation overhead; diagnoses the dominant bottleneck convincingly.	Reasoning is largely correct, but not always fully evidenced or consistently tied to the observed timings.	Attempts a bottleneck discussion, but the classification is simplistic, weakly justified, or partially incorrect.	Misclassifies bottlenecks, shows major conceptual errors, or offers little real analysis.
5. Optimization Attempts	18%	Optimisations are hypothesis-driven, carefully implemented, and honestly evaluated;	At least one worthwhile optimisation is evaluated, but the reasoning or	Some tuning or optimisation is reported, but the logic is underdeveloped	Little meaningful optimisation work, or no credible

		failures are analysed insightfully; the final choice is well defended.	interpretation could be deeper.	or results are weakly analysed.	evidence that changes were tested and interpreted.
6. Comparison	16%	Comparison with the provided baseline is fair, well controlled, and technically insightful; results cover correctness and latency; major differences receive convincing root-cause explanations.	Comparison is useful and mostly fair, but some results or explanations need more detail or tighter control.	A comparison exists, but it is narrow, weakly evidenced, or mostly descriptive.	Comparison is absent, unfair, or too unclear to support any conclusion.
7. Conclusion and scholarly communication	10%	Conclusion synthesises the most important findings, lessons, and next steps; the report is easy to follow, visually clear, and professionally written.	Conclusion is clear and sensible, though somewhat routine; writing and structure are mostly strong.	Conclusion summarises the work but adds limited insight; presentation or writing issues start to affect clarity.	Conclusion is weak or missing; communication problems materially reduce readability or credibility.

Supplementary marking notes

- Correctness matters throughout. A faster kernel that breaks numerical correctness should not receive high marks for optimisation or comparison.
- Markers should reward honest negative results where the diagnosis is rigorous. An unsuccessful optimisation can still score highly if the reasoning is technically strong.
- Appendix material may support understanding, but core claims must be substantiated in the main report.